

Program and Branch Name: *BTech (CSE), Section F*

PBL Progress Report – I (Month 1)

Field	Details
Course Name	Artificial Intelligence
Course Code	CCSAI0301
Faculty Name	Dr. Mohd Nazim
Assignment Type	Group Assignment
Student(s) Name	Maneesh Ray, Chandan Kumar, Devanshu, Sanjit Kumar, Amresh Singh
ERP ID (I Year) / Roll No. (II Year onwards)	0261DCS006, 0261DCS003, 0261DCS016, 0261DCS013, 0261DCS005
Team/Group No.	Group 98
Problem Statement / Project Title	AI-Based Phishing Email Detection & Decision Support System
SDG(s)	SDG 9 – Industry, Innovation & Infrastructure
Reporting Month	Month 1
Date of Submission	13 September 2026

Student Declaration

I/We certify that this assignment is my/our own work, with all sources appropriately acknowledged. I/We understand that a 0% mark and further academic penalties may apply if the content is found to be plagiarised or inappropriately AI-generated.

Student Name & Signature(s): _____

Date: _____

PBL Progress Report – I

Review 1: Problem Understanding & Initial Progress

Review Coverage:

- Individual / Group Assignment: Unit/Module 1 and Unit/Module 2
- Capstone Project: Rubric Categories 1 and 2

Field	Details		
1. Problem Statement / Assignment or Project Title	AI-Based Phishing Email Detection & Decision Support System – analyses incoming emails and embedded URLs and returns a clear, explainable decision (Safe / Suspicious / Phishing) instead of a plain filter action.		
2. Problem Context & Background	Phishing exploits human trust rather than technical flaws. AI-generated emails, spoofed domains and disguised URLs make static rule-based filters less effective, so an adaptive, explainable detection layer is needed.		
3. Problem Identification / Understanding	Framed as a multi-feature classification and decision problem across email content, sender metadata and URL structure. Since attackers keep adapting to evade detection, the system is designed as an evolving decision process rather than a one-time static filter.		
4. Objectives	● Build a dataset-driven phishing email/URL detection model. ● Apply heuristic, search-based feature scoring to prioritise suspicious signals. ● Model the attacker-vs-detector interaction adversarially to anticipate evasion tactics. ● Provide an explainable decision-support output, not just a binary label. ● Achieve high recall on phishing samples while keeping false positives low.		
5. Target Users / Stakeholders	Individual email users; employees within an organisation; organisational IT/security teams; email service or gateway providers who could integrate the decision-support layer.		
6. Research / Literature Review Completed	Salahdine, El Mrabet & Kaabouch (2022), "Phishing Attacks Detection – A Machine Learning-Based Approach" Selects 10 features from 10,000+ real phishing emails and compares ML classifiers for detection — informs the search-based, heuristic feature-scoring approach used here (Unit/Module 1). Kamran, Sengupta & Tavakkoli (2021), "Semi-supervised Conditional GAN for Phishing URL Detection: A Game-Theoretic Perspective" Models the attacker and the detector as two players in a game — supports framing phishing detection as adversarial search (Unit/Module 2). A broader review of public phishing datasets and common feature sets (URL length, domain age, sender-domain mismatch, keyword patterns) also guided dataset selection; a fuller literature survey with formal citations will be compiled alongside Month 2 work.		
7. Unit/Module Concepts Identified for Application	Unit/Module	Topic/Concept	Application in Project
	Unit 1 – Problem Solving Methodologies	Informed Search & Heuristic Scoring	Heuristic function scores email features (domain mismatch, urgency keywords, suspicious links) to triage risk quickly.
	Unit 2 – Adversarial Search	Minimax & Alpha-Beta Pruning	Attacker–detector interaction modelled as a two-player game; minimax-style reasoning anticipates evasion (obfuscated links, spoofed senders), pruning keeps decisions fast.
8. Work Completed During Month 1	● Finalised and got the project topic approved. ● Formed the 5-member team and assigned roles. ● Shortlisted a public Kaggle phishing email/URL dataset as the base dataset. ● Drafted the problem statement, objectives and initial system architecture.		

	Mapped AI course Module 1 (Search) and Module 2 (Adversarial Search) concepts to the detection and decision-support design.
9. Evidence / Supporting Material	Approved Topic Approval Form; shortlisted Kaggle phishing-email/URL dataset reference; initial problem-framing notes and draft system architecture (maintained in the team's project workspace).
10. Challenges Faced	Inconsistent labelling and feature availability across public phishing datasets; aligning the adversarial-search (game-theoretic) framing with a practical, implementable pipeline; coordinating schedules across a 5-member team.
11. Individual / Team Contribution	Maneesh Ray (Team Lead & AI Solution Architect): overall project direction and system design. Chandan Kumar (Search Algorithm & Optimisation Analyst): heuristic/search-based feature scoring design. Devanshu (Heuristic & Decision Intelligence Specialist): decision-rule and risk-scoring logic. Sanjit Kumar (Knowledge Representation & Expert System Designer): knowledge/rule-base design for phishing indicators. Amresh Singh (Statistical AI Analyst & Documentation Lead): dataset statistics and Month 1 report documentation.
12. Overall Progress (%)	20% — topic approval, team formation, dataset shortlisting, problem framing and Module 1–2 concept mapping completed.
13. Plan for Next Review	Begin data pre-processing and feature engineering on the chosen dataset; implement a baseline heuristic/search-based detector; start incorporating Module 3 (Knowledge Representation) and Module 4 (Statistical Reasoning) concepts for Month 2; prepare interim results for the next review.

Faculty Evaluation:

- To be completed by Faculty in Moodle